

Object Oriented Programming & JAVA Lab (OCCEEE481A/OECEEE481A)
DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING/
DEPARTMENT OF ELECTRICAL ENGINEERING

LIST OF ASSIGNMENTS

Assignment No.	Details
Assignment 1	Write a Java program to demonstrate how to use this to call another constructor in the same class. Create a class Rectangle with two constructors: One constructor with no parameters that calls the parameterized constructor with default values. One parameterized constructor that initializes length and width. Tasks: Define the class Rectangle with two constructors. Use this() in the no-argument constructor to call the parameterized constructor. Display the values of length and width.
Assignment 2	Create a Java class Person that contains a method display() which prints the person's name. Use another method show() that uses the this keyword to call display() from within the same class. Tasks: Define the Person class with methods display() and show(). Use this.display() inside the show() method. Call show() from the main method to print the person's name.
Assignment 3	Consider a multi-level inheritance hierarchy with classes Vehicle, Car, and SportsCar. If Vehicle defines a method start(), Car overrides it, and SportsCar further overrides it, how can you call the method from SportsCar to display a message from both Car and SportsCar without using the super keyword?
Assignment 4	Create an example where a Department can add and remove multiple Employee objects dynamically, but the Employee objects can exist independently of the Department. How would you model this relationship using aggregation?
Assignment 5	Employee Bonus Calculation: Develop a Employee class with a method calculateBonus(). Extend this class into Manager and Developer, overriding calculateBonus() to return different bonus values. Instantiate objects and display their bonuses.
Assignment 6	Real-world Hierarchical Inheritance: Design an inheritance hierarchy for a University System with University → Department → Student. Implement proper method overriding and access control.